

Login and Access

- The system supports multiple roles:
 - **Student:** Can register complaints, upload evidence, and track complaint progress.
 - **Warden / Faculty:** Can view and manage complaints assigned to their domain.
 - **Admin:** Oversees all user roles, resolves escalations, and maintains audit logs.
- Users log in using their credentials (registered email or roll number).
- Passwords are stored securely using **AES encryption**.
- Login validations ensure the correct role-based dashboard is loaded.

Dashboards

Student Dashboard

- **Register Complaint:**
Allows students to report issues under various categories (listed below).
The form includes:
 - Complaint Title
 - Description
 - Category Selection
 - Evidence Upload (image or document)
- **Track Complaints:**
Displays complaint ID, date, category, and current status (Pending / In Progress / Resolved).
Students can check updates made by wardens.
- **Feedback Section:**
Students can submit feedback once a complaint is resolved.
Ratings and comments are stored for quality assessment.
- **Notifications Panel:**
Shows updates when a complaint status changes.

Warden / Faculty Dashboard

- Displays all complaints categorized by:
 - **Pending**
 - **In Progress**
 - **Resolved**
- Wardens can:
 - Assign tasks to maintenance teams.
 - Update status.
 - Upload response notes or proof of resolution.
- Faculty or administrative users can filter complaints by hostel block or category.

Admin Dashboard

- Can view all complaints across users and hostels.
- Has authority to:
 - Reassign unresolved complaints.
 - Add/remove users.
 - Generate analytics (e.g., number of complaints per month).
 - View system logs and user activity for security audits.

Complaint Categories

The system includes the following complaint categories:

1. **Electrical Issues**
 - Fan, light, or switch malfunction
 - Power cut in room or corridor
2. **Plumbing Issues**
 - Water leakage
 - Broken taps or blocked drainage
3. **Mess Issues**

- Poor food quality
- Unhygienic conditions in the mess
- Refund or billing issues
- 4. **Hygiene / Cleaning**
 - Unclean washrooms or corridors
 - Dustbins not cleared
- 5. **Furniture and Maintenance**
 - Broken bed or chair
 - Damaged windows or doors
- 6. **Internet / Wi-Fi**
 - Connectivity problems
 - Login access issues
- 7. **Harassment / Security**
 - Misbehavior, harassment, or safety concerns
 - Anonymous complaint option for privacy
- 8. **Others**
 - Any issue not covered in the above categories
 - Emergency issues like fire, medical emergency, or flooding

Student Dashboard

Headings / Cards

- *Welcome, [Student Name]*
- Profile details: Name | Register No | Hostel Block | Room No | Mobile | Email
- Statistics:
 - Total Complaints
 - Pending
 - In Progress
 - Resolved
 - Rejected

Buttons / Links

- File New Complaint
- My Complaints
- Complaint Status
- Logout

File New Complaint Form

Form fields

1. Complaint Title – text input
2. Category – dropdown list
 - Electrical Issues • Plumbing • Mess/Food • Cleanliness • Furniture/Maintenance
 - Internet/Wi-Fi • Security/Harassment • Others
3. Sub-category (dynamic)
4. Description – textarea
5. Upload Image/Evidence (optional)
6. Severity – Low / Medium / High / Emergency
7. Location (Room or Block)
8. Anonymous Mode (checkbox)
9. Declaration checkbox: “I confirm the details are true.”

Buttons

- Submit Complaint - Reset/Cancel

After submission the page displays: “Complaint ID CMP-#### generated successfully.”

My Complaints / Status Page

Table columns

- Complaint ID | Date | Category | Severity | Status | Assigned Staff | Last Updated | Actions

Status badges

-  Pending  In Progress  Resolved  Rejected

Actions

- View Details → shows full record and photo
- Feedback → 1–5 stars + remarks
- Download Report (PDF)

Faculty / Warden Dashboard

Summary cards

- Total Complaints Assigned
- Pending
- In Progress
- Resolved

Tabs / Filters

- My Reports
- Newly Assigned
- Emergency Cases

Each entry shows

- Complaint ID, Title, Category
- Student Block & Room No
- Description text
- Attachment link
- Status dropdown (Pending / In Progress / Resolved / Rejected)
- Remarks box
- Update Status button

Admin Panel (visible on later screens)

Functions

- View All Complaints (filter by block, status, date)
- Assign Warden (automatic or manual)
- Analytics Dashboard – category-wise bar and pie charts
- User Management – Add / Remove Student or Faculty accounts
- Activity Logs and Audit Trail
- Export Reports to CSV or PDF

Chatbot Assistant Widget

- Appears bottom-right on all pages.
- Greets user: “Hi! How can I help you today?”
- Handles FAQ topics:
 - How to file a complaint
 - Track complaint status
 - Meaning of each status
 - Editing or deleting a complaint
 - Emergency contacts
- Keyword triggers for “fire”, “electric shock”, “harassment” → shows emergency numbers and alerts warden.

Contact / Support Info (footer)

- Hostel Office : +91 98765 43210
- Helpdesk Email : hostel.helpdesk@college.edu.in
- Emergency Warden : +91 90000 11122
- Working hours : 8 AM – 8 PM

User Roles

- **Student** – Can submit complaints, view complaint status, upload evidence.
- **Faculty** – Can file disciplinary or facility-related reports.
- **Warden/Staff** – Can view, assign, and resolve complaints.
- **Admin** – Has complete visibility and access control for user management.

Complaint Filing (Faculty/Student Portal)

Common Fields:

- **Created At** – Auto-filled with current date and time.
- **Filed By** – Auto-filled (Student/Faculty).
- **Category** – Dropdown for selecting main issue type.
- **Subcategory** – Dropdown depending on the selected category.
- **Priority** – Dropdown (Low, Medium, High, Critical).
- **Date & Time of Incident** – User-specified date and time.
- **Report Content** – Includes title, description, and evidence upload.

Complaint Categories and Subcategories

Maintenance

Subcategories:

- Electricity
- Water
- Plumbing
- AC
- Wi-Fi
- Other

Common Use Cases:

- Power outage, AC malfunction, Wi-Fi disconnection, water leakage, etc.

Safety & Discipline

Subcategories:

- Security
- Harassment
- Ragging
- Theft
- Violence
- Suspicious Activity

Common Use Cases:

- Reporting student misconduct, ragging incidents, or safety threats.

Food & Mess

Subcategories:

- Quality
- Hygiene
- Quantity
- Timings
- Staff Behaviour

Common Use Cases:

- Complaints about food quality, serving timings, or staff behaviour in the mess.

Other

Subcategories:

- Misconduct

- Noise
- General

Common Use Cases:

- General grievances or minor issues not covered under other categories.

Complaint Management Workflow

1. User selects **Category** and **Subcategory**.
2. User specifies **priority** and **incident date/time**.
3. Complaint is submitted and logged into the database.
4. Concerned authority (warden/faculty/admin) gets notified.
5. Complaint status changes dynamically (Pending → In Progress → Resolved).
6. User can view complaint status in dashboard.